■ Set

lhs `=` *rhs* evaluates *rhs* and assigns the result to be the value of *lhs*. *lhs* is then replaced by *rhs* whenever it appears.

`{`l_1`,` l_2`,` ...`}` `=` `{`r_1`,` r_2`,` ...`}` evaluates the r_i, and assigns the results to be the values of the corresponding l_i.

lhs can be any expression, including a pattern. ■ `f[x_] = x^2` is a typical assignment for a pattern. Notice the presence of `_` on the left-hand side, but not the right-hand side. ■ An assignment of the form *f*`[`*args*`]` `=` *rhs* sets up a transformation rule associated with the symbol *f*. ■ Different rules associated with a particular symbol are usually placed in the order that you give them. If a new rule that you give is *more general* than existing rules, it is, however, placed after them. One rule is considered more general than another if its left-hand side is a pattern that matches the left-hand side of the other. When the rules are used, they are tested in order. ■ New assignments with identical *lhs* overwrite old ones. ■ You can see all the assignments associated with a symbol *f* using `?`*f* or `Definition[`*f*`]`. ■ If you make assignments for functions that have attributes like `Flat` and `Orderless`, you must make sure to set these attributes before you make assignments for the functions. ■ `Set` has attribute `HoldFirst`. ■ If *lhs* is of the form *f*`[`*args*`]`, then *args* are evaluated. In addition, the flattening and sorting associated with attributes `Flat` and `Orderless` on *f* are done. ■ There are some special functions for which an assignment to *s*`[`*f*`[`*args*`]]` is automatically associated with *f* rather than *s*. These functions are: `N`, `Format`, `Options`, `Default`. ■ See page 574. ■ See also: `TagSet`, `Unset`, `Clear`, `Literal`.

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